Matrices & Matrix Algebra

ANSWER KEY

Matrix addition and scalar multiplication

1 Let A have rows $\langle 1,2\rangle, \langle 3,4\rangle$ and B have rows $\langle 0,-1\rangle, \langle 2,1\rangle$. Find $A+B$ and $2A-B$.

$A+B$ has rows $\langle 1,1\rangle, \langle 5,5\rangle$. $2A$ has rows $\langle 2,4\rangle, \langle 6,8\rangle$, so $2A-B$ has rows $\langle 2,5\rangle, \langle 4,7\rangle$.

Matrix multiplication

2 Let A have rows $\langle 1,2\rangle, \langle 3,4\rangle$ and B have rows $\langle 2,0\rangle, \langle 1,3\rangle$. Find AB.

Row 1 of AB: $\langle 1(2)+2(1),\ 1(0)+2(3)\rangle = \langle 4,6\rangle$. Row 2: $\langle 3(2)+4(1),\ 3(0)+4(3)\rangle = \langle 10,12\rangle$. So AB has rows $\langle 4,6\rangle, \langle 10,12\rangle$.

Matrix multiplication is not commutative

3 For A, B above, compute BA and compare it to AB.

Row 1 of BA: $\langle 2(1)+0(3),\ 2(2)+0(4)\rangle = \langle 2,4\rangle$. Row 2: $\langle 1(1)+3(3),\ 1(2)+3(4)\rangle = \langle 10,14\rangle$. So BA has rows $\langle 2,4\rangle, \langle 10,14\rangle$ — different from AB's rows $\langle 4,6\rangle, \langle 10,12\rangle$, confirming matrix multiplication is not commutative in general.

The identity matrix and matrix inverses

4 Describe the 2×2 and 3×3 identity matrices by their rows.

I_2: rows $\langle 1,0\rangle, \langle 0,1\rangle$. I_3: rows $\langle 1,0,0\rangle, \langle 0,1,0\rangle, \langle 0,0,1\rangle$ — a 1 down the main diagonal, 0 everywhere else.

Invertibility of 2x2 matrices

5 For A with rows $\langle a,b\rangle, \langle c,d\rangle$, state the formula for A^{-1}.

A^{-1} is $\dfrac{1}{ad-bc}$ times the matrix with rows $\langle d,-b\rangle$ and $\langle -c,a\rangle$, provided $ad-bc\neq 0$.

6 Find the inverse of A with rows $\langle 3,1\rangle, \langle 2,1\rangle$.

$\det A = 3(1)-1(2) = 1$. A^{-1} has rows $\langle 1,-1\rangle, \langle -2,3\rangle$. (Check: multiplying A by this gives I_2.)

Inverse of a product

7 State the formula for $(AB)^{-1}$ in terms of A^{-1} and B^{-1}.

$(AB)^{-1} = B^{-1}A^{-1}$ (note the reversed order).

8 If A^{-1} has rows $\langle 2, 0\rangle, \langle 1, 1\rangle$ and B^{-1} has rows $\langle 1, 1\rangle, \langle 0, 1\rangle$, find $(AB)^{-1}$.

$(AB)^{-1} = B^{-1}A^{-1}$. Row 1: $\langle 1(2) + 1(1),\ 1(0) + 1(1)\rangle = \langle 3, 1\rangle$. Row 2: $\langle 0(2) + 1(1),\ 0(0) + 1(1)\rangle = \langle 1, 1\rangle$. So $(AB)^{-1}$ has rows $\langle 3, 1\rangle, \langle 1, 1\rangle$.